Mateiio / GenerativeShift

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

main

GenerativeShift / presentations / ABRHSResearchClub / ImageProcessingForDataScience_Workshop_ABRHS_ResearchClub.ipynb

Go to file t

Mateiio moved research club data science track presentation to its own folder 9ba9244 · last year

8.74 MB

Hands-on introduction to image processing for data science and deep learning.

Author and Instructor: **Matei Iordanescu**

Concepts:

1. What is an image?
2. Meet Lena
3. What actually is an image?
4. The most basic coding pattern for image processing
5. Real example of image processing
6. Kernel convolution
   -1) Vertical edges
   -2) Horizontal edges
   -3) Blurring
   -4) Sharpening
   -5) Embossing
7. Edge Detection
8. Connection to DeepLearning

In [1]:

```
from IPython.core.interactiveshell import InteractiveShell
InteractiveShell.ast_node_interactivity = "all"
```

In [2]:

```
from PIL import Image
from IPython.display import display
import math
import numpy as np
```

Concept 1: What is an image?

An image starts as a file saved on a disk in a computer. To read the image, you can use any programing language and leverage any image libraries. Here we use Python and Pillow (PIL for short), a Python library. After reading an image with PIL, you get a PIL image object which has many methods and properties you can query to understand and manipulate your image. For example, you can get the size of the image using the size function or you can display it using the display function.

Back to Summary

In [3]:

```
path = "/workspace/Data/Lena_Colored.jpeg"
img = Image.open(path)
width, height=img.size
display(img)
width, height
```

Out[3]: (512, 512)

Concept 2: Meet Lena

Lenna (or Lena) is a standard test image used in the field of digital image processing, starting in 1973. It is a picture of the Swedish model Lena Forsén, shot by photographer Dwight Hooker and cropped from the centerfold of the November 1972 issue of Playboy magazine. **Who is Lena**

Back to Summary

In [4]:
```
display(img)
```

Concept 3: What actually is an image?

An image is an array of pixels, with each pixel being a tuple with three values corresponding to its respective R, G, and B color intensities, each typically ranging from 0-255. To find a certain pixel in an image using the PIL library, you give two indices (which correspond to rows and columns) to a function, just like any other image library.

Back to Summary

In [5]:

```
# guestimate of a blue-ish pixel
p = img.getpixel((450, 100))
red_intensity = p[0]
green_intensity = p[1]
blue_intensity = p[2]
red_intensity,green_intensity,blue_intensity

# guestimate of a red-ish pixel
p = img.getpixel((50, 100))
red_intensity = p[0]
green_intensity = p[1]
blue_intensity = p[2]
red_intensity,green_intensity,blue_intensity
```

Out[5]: (101, 22, 64)

Out[5]: (232, 140, 120)

Concept 4: The most basic coding pattern for image processing

The most basic pattern is two embedded loops that iterate over an image's rows and columns.

Back to Summary

In [6]:

```
def DuplicateImg(img):
    width, height=img.size
    newimg = Image.new("RGB", (width, height), "white")
    for row_counter in range(0, height):
        for column_counter in range(0, width):
            p = img.getpixel((row_counter, column_counter))
            newimg.putpixel((row_counter,column_counter),p)
    return newimg
```

In [7]:

```
DuplicateImg(img)
```

Out[7]:

Concept 5: Real example of image processing

Thresholding:

- One of the simplest example of image processing
- Here we generate a black and white image

Back to Summary

In [8]:

```
def Threshold(img, threshold):
    width, height=img.size
    newimg = Image.new("RGB", (width, height), "white")

    for row_counter in range(0, height):
        for column_counter in range(0, width):
            p = img.getpixel((row_counter-1, column_counter-1))
            r = p[0]
            g = p[1]
            b = p[2]
            if ((r + g + b)/3)>=threshold:
                newimg.putpixel((row_counter,column_counter),(255,255,255))
            else:
                newimg.putpixel((row_counter,column_counter),(0,0,0))
    return(newimg)
```

In [9]:

```
display(img)
Threshold(img, 150)
```

Out[9]:

Concept 6: Kernel convolution

In this tutorial, we will focus on kernel convolution which is a more advanced and powerful type of image-processing technique. Kernel convolution involves an N by N matrix (also called a kernel), with a matching N by N sliding window in the image.

Back to Summary

In [10]:

```
def Normalize(nparray: np.ndarray, maxv: float=255):
    nparray0 = nparray.astype(np.float64)
    width= nparray0.shape[0]
    height= nparray0.shape[1]
    min_intensity = nparray.min()
    max_intensity = nparray.max()
    newnparray = maxv* (nparray0 - min_intensity)/(max_intensity - min_intensity)
    return newnparray

def ConvolveImgKer(img, kernel):
    width, height=img.size
    newimgnparray = np.zeros(((height-2), (width-2), 3)).astype(np.float64)
    float_kernel = [[float(x) for x in sublist] for sublist in kernel]
    for row_counter in range(1, height-1):
        for column_counter in range(1, width-1):
            Gp=0
```

```
            Gp=0
            p = img.getpixel((row_counter-1, column_counter-1))
            Gp += (float(p[0] + p[1] + p[2])/3)*float_kernel[0][0]
            p = img.getpixel((row_counter-1, column_counter))
            Gp += (float(p[0] + p[1] + p[2])/3)*float_kernel[0][1]
            p = img.getpixel((row_counter-1, column_counter+1))
            Gp += float((p[0] + p[1] + p[2])/3)*float_kernel[0][2]
            p = img.getpixel((row_counter, column_counter-1))
            Gp += (float(p[0] + p[1] + p[2])/3)*float_kernel[1][0]
            p = img.getpixel((row_counter, column_counter))
            Gp += (float(p[0] + p[1] + p[2])/3)*float_kernel[1][1]
            p = img.getpixel((row_counter, column_counter+1))
            Gp += (float(p[0] + p[1] + p[2])/3)*float_kernel[1][2]
            p = img.getpixel((row_counter+1, column_counter-1))
            Gp += (float(p[0] + p[1] + p[2])/3)*float_kernel[2][0]
            p = img.getpixel((row_counter+1, column_counter))
            Gp += (float(p[0] + p[1] + p[2])/3)*float_kernel[2][1]
            p = img.getpixel((row_counter+1, column_counter+1))
            Gp += (float(p[0] + p[1] + p[2])/3)*float_kernel[2][2]
            newimgnparray[column_counter-1][row_counter-1][0]=Gp
            newimgnparray[column_counter-1][row_counter-1][1]=Gp
            newimgnparray[column_counter-1][row_counter-1][2]=Gp
    return Image.fromarray(Normalize(newimgnparray).astype(np.uint8))
```

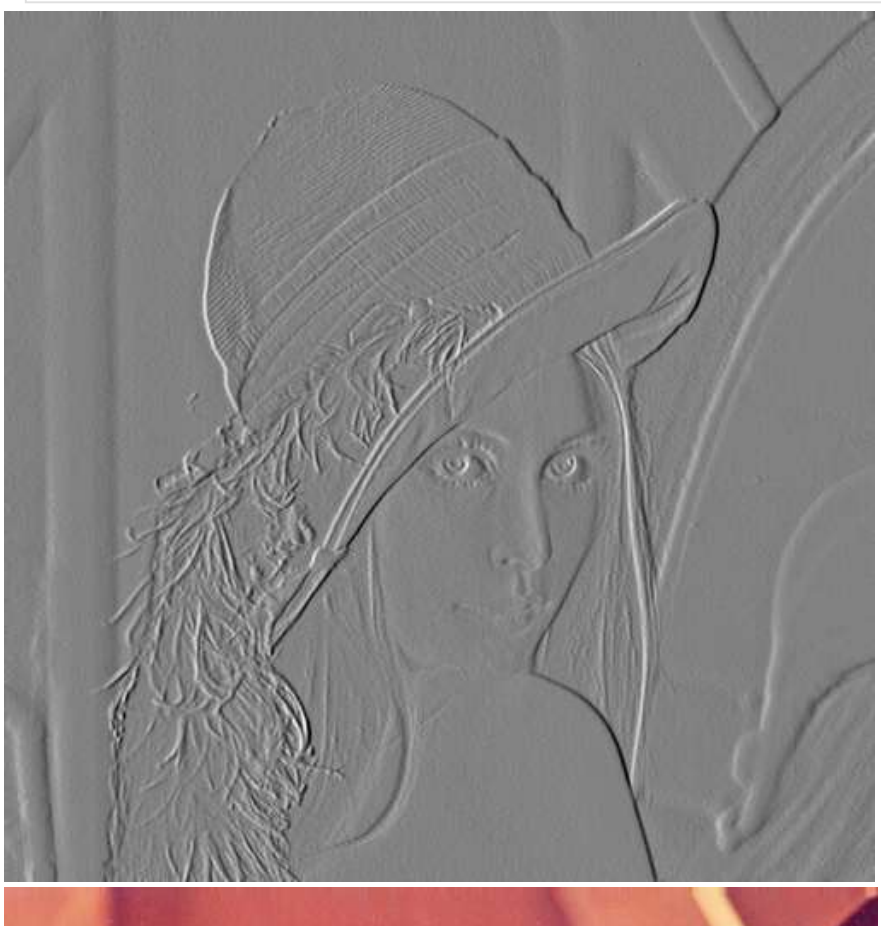
Practical examples:

1: Vertical edges

Back to Summary

In [11]:

```
kernel=[[0,-1,0],[0,0,0],[0,1,0]]
convolvedimg=ConvolveImgKer(img, kernel)
display(convolvedimg)
display(img)
```



2) Horizontal edges

Back to Summary

In [12]:

```
kernel=[[0,0,0],[-1,0,1],[0,0,0]]
convolvedimg=ConvolveImgKer(img, kernel)
display(convolvedimg)
display(img)
```

3: Blurring

Back to Summary

In [13]:

```
kernel=[[1,1,1],[1,1,1],[1,1,1]]
path = "/workspace/Data/Lena.jpeg"
img_small = Image.open(path)
convolvedimg=ConvolveImgKer(img_small, kernel)
for counter in range(10):
    convolvedimg=ConvolveImgKer(convolvedimg, kernel)
display(convolvedimg)
display(img_small)
```

4: Sharpening

Back to Summary

In [14]:

```
kernel=[[0,-1,0],[-1,5,-1],[0,-1,0]]
convolvedimg=ConvolveImgKer(img, kernel)
display(convolvedimg)
display(img)
```

5: Embossing

Back to Summary

In [15]:

```
kernel=[[-2,-1,0],[-1,1,1],[0,1,2]]
convolvedimg=ConvolveImgKer(img, kernel)
display(convolvedimg)
display(img)
```

Concept 7: Edge Detection

Leverage the gradient to detect edges at all angles

Back to Summary

In [16]:

```
def GradientMagnitude(img):
    width, height=img.size
    newimg = Image.new("RGB", (width-2, height-2), "white")
    for row_counter in range(1, height-1):
        for column_counter in range(1, width-1):
            p1 = img.getpixel((row_counter, column_counter-1))
            p2 = img.getpixel((row_counter, column_counter+1))
            Gx=((p2[0] + p2[1] + p2[2])/3)-((p1[0] + p1[1] + p1[2])/3)
            p1 = img.getpixel((row_counter-1, column_counter))
            p2 = img.getpixel((row_counter+1, column_counter))
            Gy=((p2[0] + p2[1] + p2[2])/3)-((p1[0] + p1[1] + p1[2])/3)
            magnitude=math.sqrt((Gx * Gx) + (Gy * Gy))
            magnitude = int(magnitude)
            newimg.putpixel((row_counter-1,column_counter-1),(magnitude,magnitude,magnitude))
    return newimg
```

In [17]:

```
GradientMagnitude(img)
display(img)
```

Out[17]:

Concept 8: Connection to DeepLearning

Machine Learning (ML):

- A class of algorithms that learn thresholds from input data.
- Creates models to classify new, unseen data.

Artificial Neural Networks (ANN):

- A subset of ML using artificial neurons grouped in layers.
- Performs learning tasks by mimicking brain-like processes.

Deep Learning (DL):